$\mathbb{C}$ - parity (charge conjugation)

$\mathbb{C}$: particle $\longrightarrow$ anti-particle   internal symmetry

$(-1) \times$ all quantum numbers

$\vec{s}, \vec{x}, t$ are unchanged

neutral particles potential eigenstates.

γ, π^0

$$\pi^0 \longrightarrow \gamma\gamma \qquad \checkmark \text{ allowed.}$$

$$C \quad 1 \qquad (-1)(-1) = +1.$$

$$\pi^0 \longrightarrow \gamma\gamma\gamma$$

$$C \quad +1 \qquad C = (-1)^3 = -1 \qquad \times \text{ disfavored.}$$

Experimental indication that $\mathbb{C}$ is conserved in EM, strong interaction

$\mathbb{C}$ violation weak interactions

In QFT: $C = -1$ for fermion - anti-fermion

$C = +1$ for boson - anti-boson.

neutral elementary fermion is $\nu \overset{?}{=} \bar{\nu}$

In SM we use $\nu \neq \bar{\nu}$

Multi-particle states

look at $\mathbb{C}$ conservation for states of particles.

$$\mathbb{C} \mid Q = 0 \rangle = a \mid Q = 0 \rangle$$

total charge of state $Q = 0$

$$\mathbb{C}(\pi^+\pi^-) \longrightarrow \pi^-\pi^+$$

$$\pi^+\pi^-\pi^0 \longrightarrow \pi^-\pi^+\pi^0$$

$$\underset{Q_a = 0}{a} \xrightarrow[\text{QCD.}]{\text{EM}} \underset{Q_b = 0}{b} \Rightarrow \mathbb{C}_a = \mathbb{C}_b$$

$$a \longrightarrow \pi^+\pi^-\pi^0$$

$Q_c = 0$ $\quad \pi^+\pi^-$

$\quad \pi^0\pi^0$

$C_a = ?$

__Example with spinless particles__

$\pi^+\pi^-$ $\qquad C: \pi^+\pi^- \longrightarrow \pi^-\pi^+$

$$C\,\psi_{\pi^+\pi^-} = (-)\,\psi_{\pi^+\pi^-}$$

$$\psi = \psi_{space}\,\psi_{spin}\,\psi_{intrinsic}$$

$$\pi^+ \longleftarrow \qquad \longrightarrow \pi^- \qquad\qquad \pi^+ \longleftarrow \qquad \longrightarrow \pi^-$$

$$C \Downarrow \qquad\qquad\qquad\qquad P \Downarrow$$

$$\pi^- \longleftarrow \qquad \longrightarrow \pi^+ \qquad\qquad \pi^+ \longrightarrow \qquad \longleftarrow \pi^-$$

$$\equiv$$

$$\pi^- \longleftarrow \qquad \longrightarrow \pi^+$$

$$C\,\psi_{\pi^+\pi^-} \equiv P\,\psi_{\pi^+\pi^-} = (-1)^L\,\psi_{\pi^+\pi^-}$$

L: orbital angulare mom. of $\pi^+\pi^-$ pair.

state $\pi^+\pi^-$ particle–anti-particle pair

Example: e^+e^- $\quad s = 1/2$ for each fermion. $\quad C: e^+ \to e^-$

$$e^+ \overset{\Rightarrow}{\longleftarrow} \quad \overset{\Leftarrow}{\longrightarrow} e^- \quad \overset{C}{\Longrightarrow} \quad e^- \overset{\Rightarrow}{\longleftarrow} \quad \overset{\Leftarrow}{\longrightarrow} e^+$$

$$e^+ \overset{\Rightarrow}{\longleftarrow} \quad \overset{\Leftarrow}{\longrightarrow} e^- \quad \overset{P}{\Longrightarrow} \quad e^- \overset{\Leftarrow}{\longleftarrow} \quad \overset{\Rightarrow}{\longrightarrow} e^+$$

$$e^+ \overset{\Rightarrow}{\longleftarrow} \quad \overset{\Leftarrow}{\longrightarrow} e^- \quad \underset{S}{\Longrightarrow} \quad e^+ \overset{\Leftarrow}{\longleftarrow} \quad \overset{\Rightarrow}{\longrightarrow} e^-$$

spin flip

$$C \equiv P \times S$$

Same effect.

$$e^+ \overset{\Rightarrow}{\longleftarrow} \quad \overset{\Leftarrow}{\longrightarrow} e^- \quad \overset{P \times S}{\Longrightarrow} \quad e^- \overset{\Rightarrow}{\longleftarrow} \quad \overset{\Leftarrow}{\longrightarrow} e^+$$

$\mathbb{P}$: $(-1)^L$

$\$$: $(-1)^{S+1}$

Recall our spin flip.

$$\$ \psi_{F\bar{F}} = (-1)^{S+1} \psi_{F\bar{F}}$$

$$\$ \psi_{B\bar{B}} = (-1)^{S} \psi_{B\bar{B}}$$

$$\mathbb{C} \psi_{e^+e^-} \equiv (\mathbb{P} \times \$) \psi_{e^+e^-} = (-1)^L (-1)^{S+1} \underbrace{C_{e^+e^-}}_{\text{from QFT } (-1)_{F\bar{F}}} \psi_{e^+e^-}$$

e^-e^+ : has $\mathbb{C}$ parity of $(-1)^{L+S+1+1}$

$(-1)^{L+S}$

In general for particle - anti-particle state $(p\bar{p})$

$$\psi_{p\bar{p}} = \psi_{space} \psi_{spin} \psi_{intrinsic}$$

Fermions.

$$\mathbb{P} \psi_{pp} = (-1)^L \psi_{space}.$$

$$\$ \psi_{pp} = (-1)^{S+1} \psi_{pp}$$

$$\mathbb{C}^{F\bar{F}}_{intrinsic} = (-1)$$

$$\mathbb{C} \psi_{F\bar{F}} = (-1)^{L+S+1+1} = (-1)^{L+S}$$

Bosons:

$$\mathbb{P} \psi_{B\bar{B}} = (-1)^L \psi_{B\bar{B}}$$

$$\$ \psi_{B\bar{B}} = (-1)^S \psi_{B\bar{B}}$$

$$\mathbb{C}^{intrinsic}_{B\bar{B}} = +1. \quad \text{(from QFT)}$$

$$\mathbb{C} \psi_{B\bar{B}} = (-1)^{L+S}$$

$p\bar{p}$ state so $Q = 0$ $\Rightarrow$ $\mathbb{C}_{p\bar{p}} = (-1)^{L+S}$

Exercise:

$p\bar{p}\ \pi^+\pi^-$

$\underbrace{p\bar{p}}\ pp$

$\pi^+\pi^-\pi^0$

$\pi^0\pi^0\pi^0$

$\pi^+\pi^-\pi^+\pi^-$

G-parity

very few $\mathbb{C}$ eigenstates.

Think about I multiplet π^+ π^0 π^-

$|\pi^+\rangle = |1,+1\rangle$ $|\pi^0\rangle = |1,0\rangle$ $|\pi^-\rangle = |1,-1\rangle$

$\mathbb{C}: \pi^+ \to \pi^-$ $I_3 = +1 \to I_3 = -1$

$$\mathbb{G} = \mathbb{C} \times \mathbb{R}_{I_2} = \mathbb{C}\, e^{i\pi I_2}$$

$\mathbb{C}: \pi^+ \to \pi^-$

$\mathbb{R}_2: \pi^- \to \pi^+$

$\mathbb{G}: |\pi^+\rangle \to A|\pi^+\rangle$

$\mathbb{C}|\pi^+\rangle = \alpha|\pi^-\rangle$

$\mathbb{R}_2|\pi^-\rangle = e^{i\pi I_2}|\pi^-\rangle = (-1)^{I-I_3}|\pi^+\rangle$

$|\pi^+\rangle = |I=1, I_3=+1\rangle$

$$\mathbb{G}|\pi^0\rangle = \mathbb{C}\,\mathbb{R}_2|\pi^0\rangle = \underbrace{C_{\pi^0}}_{+1}(-1)^{I-I_3}|\pi^0\rangle$$

$$= (+1)(-1)^{1-\phi}|\pi^0\rangle$$

$$= -1\,|\pi^0\rangle$$

$G = -1$ for all multiplet members:

$G|\pi^+\rangle = -|\pi^+\rangle = (-1)^I|\pi^+\rangle$

$G|\pi^0\rangle = -|\pi^0\rangle = (-1)^I|\pi^0\rangle$

$G|\pi^-\rangle = -|\pi^-\rangle = (-1)^I|\pi^-\rangle$

Only for multiplets wit $\sum Q_i = 0$

I = integer

example with $I = 1/2$ $\begin{pmatrix} K^+ \\ K^0 \end{pmatrix}$ $\mathbb{C} \to K^+ \to K^-$

$\mathbb{R}_2 K^- \to \bar{K}^0 \neq K^0$

G-parity eigenvalue

$\mathbb{G} = \mathbb{C} \times \mathbb{R}_2 \longrightarrow G = \underbrace{(-1)^{L+S}} (-1)^{I} = (-1)^{L+S+I}$

pion I-multiplet: $L=0$ ground state

$S=0$ bosons with $S=0$

$I=1$

G-parity conserved in QCD, EM

$\boxed{\omega(782)}$ $I^G(J^{PC}) = 0^-(1^{--})$

Mass = 782 MeV

$G_\omega = -1$

$G_\pi = -1$

ω(782) DECAY MODES

	Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1	$\pi^+\pi^-\pi^0$	(89.2 ± 0.7) %	
Γ_2	$\pi^0\gamma$	(8.35 ± 0.27) %	S=2.2
Γ_3	$\pi^+\pi^-$	(1.53 ± 0.12) %	S=1.2
Γ_4	neutrals (excluding $\pi^0\gamma$)	$(7\ {}^{+8}_{-4}) \times 10^{-3}$	S=1.1
Γ_5	$\eta\gamma$	$(4.5 \pm0.4) \times 10^{-4}$	S=1.1
Γ_6	$\pi^0 e^+ e^-$	$(7.7 \pm0.6) \times 10^{-4}$	
Γ_7	$\pi^0\mu^+\mu^-$	$(1.34\pm0.18) \times 10^{-4}$	S=1.5
Γ_8	$\eta e^+ e^-$		
Γ_9	$e^+ e^-$	$(7.38\pm0.22) \times 10^{-5}$	S=1.9
Γ_{10}	$\pi^+\pi^-\pi^0\pi^0$	$< 2 \times 10^{-4}$	CL=90%
Γ_{11}	$\pi^+\pi^-\gamma$	$< 3.6 \times 10^{-3}$	CL=95%
Γ_{12}	$\pi^+\pi^-\pi^+\pi^-$	$< 1 \times 10^{-3}$	CL=90%

$\omega \rightarrow \pi^0\pi^-\pi^+$

$G \quad -1 \qquad (-1)^3$

$\omega \longrightarrow \pi^+\pi^-$

$G=-1 \qquad G=+1$

$Q(\omega\to 3\pi) < Q(\omega \to 2\pi)$.

$782 - 280 = 500$ MeV.

$(782 - 3\times140)$ MeV $= 360$ MeV

I was exact symm $\quad M_n \neq m_p \quad I = 1/2 \quad \begin{pmatrix} n \\ p \end{pmatrix}$

$I =$ integer $\Rightarrow \sum_i Q_i = 0$ for multiplet.

elements $2I+1$ states. $\quad \sum_i I_3^i = 0$.

$I = 1/2$ (half integer)

$2I+1$ states $\quad \begin{pmatrix} n \\ p \end{pmatrix}$

electric charge
$Q \longleftrightarrow$ Isospin.

Nucleus of A nucleons with Z protons. $^{A}_{Z}N$

$$I_3 = Z\, I_3^{p} + (A-Z)\, I_3^{n}$$

$$= +\frac{Z}{2} + (A-Z)(-1/2) = \frac{Z}{2} - \frac{A}{2} + \frac{Z}{2} =$$

$$= Z - \frac{A}{2}$$

$$\Rightarrow \quad Z = I_3 + \frac{A}{2}$$

$$\downarrow$$

$$Q_e = I_3 + \frac{A}{2}$$

$A \to$ # baryons.

Gell-Mann - Nishijima Formula.

$$Q = I_3 + \frac{B}{2}$$

p: $Q = \frac{1}{2} + \frac{1}{2} = +1$ n: $Q = -\frac{1}{2} + \frac{1}{2} = 0$

pions: π^+ $Q = +1 + \frac{0}{2} = +1$

Strange mesons: $K^+, K^-, K^0, \bar{K}^0$ Strange particles.

$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix}$ flavor quantum numbers for hadrons.

$$Q = I_3 + \frac{B + S + C + b + t}{2}$$

$\begin{pmatrix} u \\ d \end{pmatrix} = I = 1/2$ doublet

c, s, b, t singlet $I = 0$

$B^0 = \bar{b} d$

$$Q = I_3 + \frac{b=1}{2} = -\frac{1}{2} + \frac{1}{2} = 0.$$

Mesons: $q_i \bar{q}_j$ $(i,j = u, d, s, c, b)$

Baryons: $q_i q_j q_e$

Static quark model with SU(3) flavor for classification of hadrons.

Name	$\pi^{\pm}$	π^0	$K^{\pm}$	K^0	η	p	n	Λ	$\Sigma^{\pm,0}$	Δ
Mass (MeV)	140	135	494	498	548	938	940	1116	1190	1232
Charge	±1	0	±1	0	0	1	0	0	±1,0	2,±1,0
Parity	–	–	–	–	–	+	+	+	+	+
Baryon n.	0	0	0	0	0	1	1	1	1	1
Spin	0	0	0	0	0	½	½	½	½	$^3/_2$

static model proposed by Gell-Man Zwig 1961–1964
Eight fold way

how to extend Isospin to include strangeness.

Isospin SU(2) symmetry

to extend to SU(N) symm. to incorporate strangeness.

SU(N) : special unitary operator group
det = 1 trace = 0

$U \in SU(N)$ – $U = e^{i a_j T_j}$ T_j : generators

generators = $N^2 - 1$ with $N-1$ diagonal operators

SU(2) : $2-1 = 1$ diagonal operator.
Spin : S_z, S_3 is the diagonal operator
isospin: I_3 is the diagonal operator.

Consider SU(3)
2 diagonal operators: I_3, S (strangeness).

explain mass structure of all hadrons include strange ones.

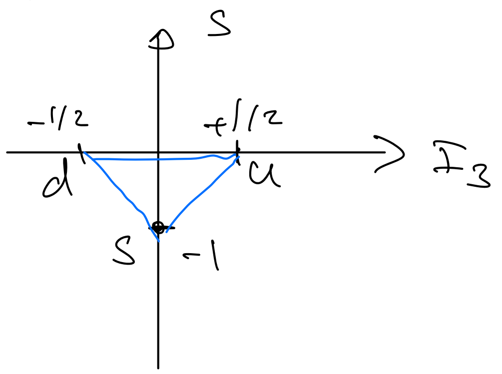
up
down $\begin{pmatrix} u \\ d \end{pmatrix} = \begin{pmatrix} +1/2 \\ -1/2 \end{pmatrix}$ isospin doublet with $I = 1/2$ S = 0 for u, d

strange s I = 0 isospin singlet strangeness S = −1 for s quark

u, d, s: 3 elements of the fundamental representation of SU(3)

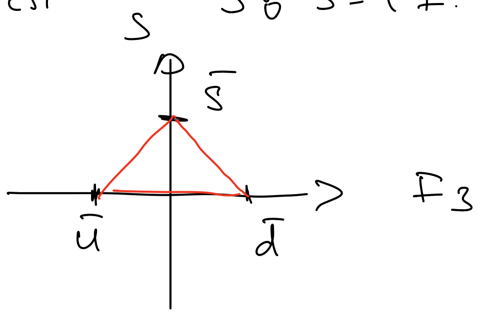
$u = \begin{pmatrix}1\\0\\0\end{pmatrix}$ $d = \begin{pmatrix}0\\1\\0\end{pmatrix}$ $s = \begin{pmatrix}0\\0\\1\end{pmatrix}$ Fundam. representation

3



anti-quarks:
$\bar{u} \Rightarrow I_3 = -1/2$
$\bar{d} = I_3 = +1/2$
$\bar{s}$ $I_3 = 0$

Strangeness $\bar{s}$: $S = +1$. $\bar{u}, \bar{d}$: $S = 0$.



For quarks: $B = 1/3$ Baryon number

$\bar{q}$: $B = -1/3$

$$Q = I_3 + \frac{B + S}{2}$$

up: $Q_u = +\frac{1}{2} + \frac{1}{2}\,\frac{1}{3} = \frac{1}{2} + \frac{1}{6}$
$= \frac{4}{6} = \frac{2}{3}$

fractionally charged particle

Mesons: $q_i \bar{q}_j$ $\underset{\substack{q_i\\ \text{quarks}}}{3} \otimes \underset{\substack{\bar{q}_j\\ \text{anti-quarks}}}{\bar{3}} = 1 \oplus 8$ in $SU(3)$.

Quarks: Fundamental representation 3 of $SU(3)$
anti-quarks: Fundament. Conjugate representation $\bar{3}$ of $SU(3)$

Mesons: multiplets of $SU(3)$ as a result of product between quarks and anti-quarks $(3 \otimes \bar{3})$

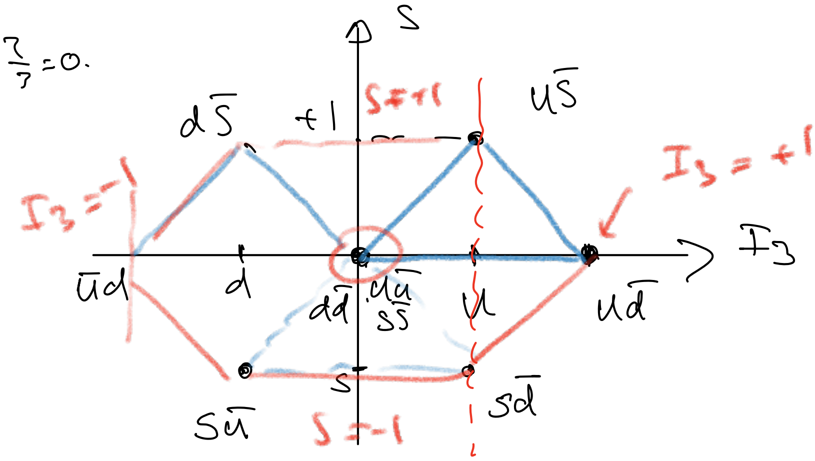
$u\bar{u}$: $I_3 = \frac{2}{3} - \frac{2}{3} = 0.$

$u\bar{s}$

$u\bar{d}$

$I_3(u) = +1/2$

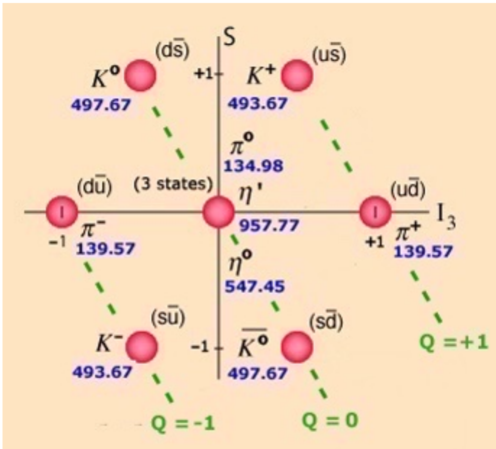
$I_3(\bar{d}) = +1/2$



$u\bar{u}, d\bar{d}, s\bar{s}$: $I_3 = 0$ $S = 0.$

Assume same mass for u, d, s

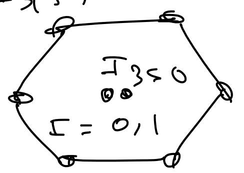
$u\bar{s}, s\bar{d}$: $I_3 = +1/2$



SU(2)

$\frac{1}{2} \otimes \frac{1}{2} = 1 + 3$

(I_3, S)

$3 \otimes \bar{3} =$  $+$ $\bullet$ $I_3 = 0$, $S = 0$

8 octet $\oplus$ 1 singlet

$\pi^+ = u\bar{d}$ — — vertices of octet are simple

States with $I_3 = 0$, $S = 0$

$u\bar{u}, d\bar{d}, s\bar{s}$ combinations $\longleftrightarrow$ π^0, η, η' physical states

octet : symmetric under flavor exchange ($u \leftrightarrow d$, $u \leftrightarrow s$, $d \leftrightarrow s$)

singlet : antisymm. under flavor exchange